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ARTICLE



Industry 4.0 and the future of quality work in the global digital economy

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ABSTRACT

This paper engages with debates in academic literatures over Industry 4.0 (i4.0) and the Future of Work (FoW), critiquing their development in practical isolation from each other despite i4.0 recently coming to resemble in policymaking a widely adopted narrative for the FoW in the digital age. We argue that beyond their own individual inadequacies, both approaches have common failings. We conceptualise the role of i4.0 in shaping the FoW in terms of the development of global value chains and i4.0's implications for their digitalisation, where there is an underdeveloped analysis of change. We argue that the clearest expression of i4.0 is in the impact of 'platform capitalism', which already has implications for workers in the Global South and which, via i4.0's digital integration with production, will potentially have significant implications for the quality of work in the Global North.

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Introduction

We stand on the brink of a technological revolution that will fundamentally alter the way we live, work, and relate to one another. In its scale, scope, and complexity, the transformation will be unlike anything humankind has experienced before. We do not yet know just how it will unfold, but one thing is clear: the response to it must be integrated and comprehensive, involving all stakeholders of the global polity, from the public and private sectors to academia and civil society (Schwab 2017).

This paper is concerned with innovation and the Future of Work (FoW). Specifically, it engages with two current debates, the first over Industry 4.0 (i4.0) and the way that industry, the state and economies are supposedly adapting to what are, ostensibly, the realities of the global economy. The second is, more specifically, concerned with the FoW and the discourse that has grown rapidly in media, business and academic circles and yet has not accurately reflected global economic realities in its blueprint for a sustainable future in which human labour is enhanced with the introduction of digital technologies. The two bodies of work that have developed around these topics exist in theoretical and practical isolation from each other. In this paper, we examine i4.0 in the light of insights to be drawn from the literature focused on the FoW and suggest that beyond their own

individual inadequacies both approaches have common failings. These include an inadequate approach to the analysis of development and change in the labour process; an underdeveloped analysis of transformation and change in global value chains/global production networks (GVCs/GPNs); and a decontextualisation that ignores or underplays the implications of economic crisis and environmental crisis.

We stress from the outset that this paper details i4.0 at great length because it is a less well-known concept in the Australian literature and context than is the FoW debate. Thus, although we make reference to the Australian experience of the FoW and i4.0 to provide some practical, technical and policy context, we are primarily concerned with the two theoretical frameworks. How these frameworks are rolling out in practice, and most importantly, the reaction of workers and their organisations, will be the subject of a following paper (but see Moore et al. 2018). As Mason (2017, 183) has argued, 'The real history of work cannot be written as "economics plus technology": it involves the interaction of technology with organizations created by workers, and it involves the creation of power relationships based on age, gender and ethnicity.' We begin the paper with a critical examination of the concept of innovation, before moving to i4.0 and the FoW. The bulk of material on i4.0 is derived from reports prepared by consultants, think tanks and government agencies, which are all largely uncritical, as argued by Morgan (2019), and in many cases present 'digital disruption' in a purely positive light (Dean and Spoehr 2018). This presents a gap in knowledge that we address with a more critical examination of i4.0 and the FoW. We conclude by outlining an approach to 'Platform Capitalism' with the lessons that can be learned from i4.0 and FoW.

Conceptualising innovation for the future of work

Innovation is a poorly defined concept. The most generally accepted definition of innovation comes from the OECD's Oslo Manual (OECD and Eurostat 2005) which it identifies as the implementation of a new or significantly improved product (good or service), or process; a new marketing method; or a new organisational method in business practices, workplace organisation or external relation. Yet the meaning of the term varies considerably depending on the context in which it is used, and for what purpose. Most widely used in reference to technological and scientific advancements, innovation exists along a number of axes ranging from radical (or disruptive) to incremental, first-in-organisation to first-in-world, product to process, sector to sector, as well as over the lifecycle, and simply a change in focus over time (i.e. design thinking, open innovation). One danger is that innovation is simply reduced to a notion of change. As writers from Marx and Schumpeter to more recently Stan Metcalfe and academics at the University of Sussex's Science Policy Research Unit (SPRU) have all pointed out, change is endemic to capitalism, as it is with all human organising systems. The problem is that innovation threatens to become a 'chaotic concept' (Sayer and Walker 1992), one that is politically important but analytically vacuous. It is a noisy term, so packed with different and often contradictory elements that it ceases to have any meaning.

How can we make sense of innovation as a coherent concept? Any discussion of innovation must move beyond tautology. In a world economy increasingly dominated by GVCs, businesses that participate in them are more innovative and engaged in research and development (R&D) and skills development. Such participation also challenges organisations to

upgrade their management and financial practices and technology, improving collaboration and altogether underscoring a 'step change' in business culture that helps them maintain competitiveness. A ground-breaking Harvard study reflects the benefits of this approach. *The Atlas of Economic Complexity* (Hausmann et al. 2013) measures the productive knowledge of 133 nations, translating data on tacit productive knowledge, or 'know-how' into a guide of how effectively each nation is able to mobilise and combine the knowledge and skills within its economy to produce and export a diversity of high-value products. In their 2013 results, Japan, Korea, most Western European nations and the United States all ranked in the top 20 economically complex economies. A step change driving business innovation does not occur in a vacuum. There is policy context helping to drive transformation in these high-performing countries, all of which coalesce their respective innovation strategies around digital transformation in a 'fourth industrial revolution' (4IR), commonly known as 'Industry 4.0.'

Industry 4.0: what is it?

Modern history has so far witnessed three industrial revolutions. The first saw the application of water and steam power to mechanical production processes. The second occurred when mechanical production was standardised on a mass manufacturing scale with the aid of electricity and the division of labour. The third industrial revolution involved the application of information communications technology (ICT) and electronics to the customisation and specialisation of manufacturing. All of these revolutions signified enormous disruptions to the organisation of production processes and great changes to established cultural norms and social practices. Within the third period of industrial development, the internet was one of the key innovations to emerge and fundamentally change how information is made, stored and shared. Over time, the internet has itself undergone innovations to have significant impact on machines. The development of earlier industrial systems has spanned decades but the impact of the internet and related ICT technologies have enabled change to occur at a much more rapid pace, particularly where an 'Internet of Things' (IoT) is characterised by the embedding of the internet in ICT technologies, producing a ubiquity of machines in our daily lives.

The digital connectivity enabled by the internet has created new possibilities for innovative manufacturing processes and a so-called 4IR. The technological thrust of this 4IR is what is meant by Industry 4.0 (i4.0). At the broadest level, i4.0 is the digitally driven integration of manufacturing production methods in an 'Industrial Internet of Things' (IIoT). Advanced technologies including robotics automation, sensors, artificial intelligence (AI), big data, cloud computing, virtual reality (VR), augmented reality (AR) and additive manufacturing (i.e. 3D-printing) are applied to production processes through digitalisation, linking them seamlessly to human processes in a 'cyber-physical production system' (CPPS) that is networked across cyberspace. In i4.0 production, smart machinery controls production processes by itself. These machines no longer simply process products, they actively communicate with the product and other smart machines to receive and transmit instructions relating to manufacturing procedures and are able to adjust in real-time to flexible production demands.

As a result, manufacturing processes are on the cusp of another major disruption. The digital innovations propelled by i4.0 have significant implications for production processes, supply chains and value chains. Industry 4.0 represents a transformation of

the way manufacturing firms do business and a significant opportunity to apply innovation to firms, industries and economies. In terms of implementation, i4.0 depends upon the development of 'smart factories.' These are Industry 4.0-enabled facilities that utilise augmented reality, sensors, wearables, the IoT and other physical-to-digital technologies for tracking of production, monitoring of quality control and management of the tooling life-cycle (Federal Ministry for Economic Affairs and Energy 2017). This enhances the effectiveness of digitally enabled capabilities, synchronising processes into an intelligent cyber-physical system (Sniderman et al. 2016). The outcome is industrial production supported by sensors and microchips that link the products themselves into the CPPS (Lu 2017). Such connectivity provides possibilities for an end-to-end process of communication: from the digital conceptualisation of a product, to its physical production, to the product's digital link back to the CPPS to relay critical data about the product's lifecycle, which potentially improves product development, manufacturing processes and customer satisfaction.

Industry 4.0-ready companies are those that have adopted an industrial CPPS platform and so embrace the digital organisation of their production processes. The utilisation of the smart factory by firms means creating a network that incorporates digitally connected machines, warehousing systems and production facilities into a digitalised environment with end-to-end ICT integrating all facets of manufacturing, from inbound logistics to production, marketing, outbound logistics and services (Kagermann et al. 2013). This fusion of technical processes and business processes can provide real-time quality, resource, and cost advantages when compared with typical production systems.

But perhaps even more important is the opportunity seen in i4.0 to transform work and society, as it is a system that potentially represents more than simply smart machines making things. Rather, the ability of machines to bridge the physical and virtual worlds is intrinsic to the innovative transformation of economies and societies attainable with the rapid digitalisation of production. The digitalisation of manufacturing production defines the entire life cycle of a product. Companies can develop new business models that offer complementary tertiary services to autonomous manufacturing processes, and shape key business objectives that improve their positions in the value chain (Sniderman et al. 2016). Industry 4.0 processes can facilitate value-creation to include all lifecycle phases of a product. The intelligent processes and flexible production systems that characterise i4.0 present opportunities for the creation of business models that employ smart factories to deal with mass customisation as markets for consumer goods *expand* on the basis of size but *fragment* on the basis of individualised tastes.

However, the digitalisation of a product's journey from conceptualisation and development, production, use and maintenance through to the product's recycling (Platform Industrie 4.0 2017) has the potential to impact more than merely manufacturing. Industry 4.0's supposedly seamless integration of cyber and physical worlds can mean a much broader transformation of, and impact on, society, environment and economy (for an extended critique see Zuboff 2019). Thus, when understood at the macro-scale i4.0 presents business model innovation opportunities but also potential for an innovative reorganisation of whole social and economic systems.

Key organisational components of industry 4.0

There is considerable clarity around what i4.0 can supposedly achieve for digital technology-driven changes to manufacturing production. Yet so far, no consensus has emerged on an adopted definition of i4.0 in both academic literature and in policymaking that could drive these changes in ways that are socially and economically transformative at a global level. This means that not only do interpretations vary, but different visions of the FoW are emerging. In Germany, the IoT is the key enabling platform for a far deeper transformation of manufacturing processes, which has implications for all sectors of the economy (Kagermann and Lukas 2011). The IoT is merely the driving force for the development of an IIoT which becomes a digital platform for manufacturing transformation. In comparison, the United States perspective is a view of the IoT as the fundamental component of manufacturing transformation (Kagermann et al. 2016). Thus, in the US context R&D, institution-building and policymaking are centred on the IIoT without a broader view of societal and governmental adaptations to go along with it.

The present overview takes the German 'Industrie 4.0' to most accurately encapsulate the potential impact of the digital transformation of manufacturing, where it may serve as the foundation for an industrial revolution that fundamentally alters civil, social, governmental and legal structures. Yet even the objectives of the German policy framework should be scrutinised. Fuchs (2018) argues that i4.0 is an ideology, specifically a German Ideology that promises growth in the German context. Echoing this perspective, Schroeder (2016) has theorised the evolutionary perspective inherent to this German ideology that emphasises mitigating disruption through social planning. Fuchs argues that the idea of a technological revolution comes before the actual technological and economic developments. In effect, Industry 4.0 is an attempt to talk a new technological paradigm ideologically into existence (see Zuboff 2019; Morgan 2019 for the tyranny of inevitability). However, this raises a further point which appears to be ignored in much of the i4.0 debate. There has been a growing and influential literature which questions whether policies emerging in one political-economic domain are directly and unproblematically applicable in another (e.g. Peck and Theodore 2015). This does not appear to be a problem as far as promoters of i4.0 are concerned.

The Australian literature is entirely uncritical, reflecting a passive acceptance of i4.0 ideology. The federal government has fostered an approach to Industry 4.0 through partnership with Germany and its Platform Industrie 4.0 public-private policy platform. The Australian Prime Minister's Industry 4.0 Taskforce has aided international digital collaboration and Australian digital-networked integration with Germany. The Taskforce established a partnership to cooperate and share information through five working groups related to (1) digital interoperability standards, (2) industrial transformation support for Small- and Medium-sized Enterprises (SMEs), (3) Industry 4.0 testlabs, (4) cybersecurity and work, (5) education and training needs (DIIS 2017). To date, only two reports have been produced: one recommending the adoption of Germany's digital network standards (Aljukic 2017); and a second on Industry 4.0 testlab developments in Australia (Gallagher 2017). The remaining groups have yet to contribute anything substantial in their respective areas of focus. Meanwhile, the *National Science and Innovation Agenda* (Australian Government 2015) and its prioritisation of a digitalisation strategy in which the i4.0 Taskforce sits has disappeared in the wake of former Prime Minister Malcolm

Turnbull's ousting by his party, and apparently so to have the i4.0 developments that were pursued within the broad innovation framework set out by the *Agenda* (see Green 2019).

Insofar as it has been introduced at all, i4.0 in Australia is in its very early days. It has been argued that whilst some Australian manufacturing businesses have begun an i4.0 transformation and are seeing the benefits catalysed by the 4IR, the majority are only starting to understand how they can integrate these technologies into their business and transform their workforce to engage in new ways of working. According to consultant reports, Australian manufacturing businesses urgently need a 'call to action' to understand why they must invest more in their workforces, research and technology to thrive in the next industrial revolution. Herein we see the de-contextualisation of the FoW in action: despite these calls, the Australian business literature remains apolitical despite an identifiable concern that Australia will fall behind if it does not invest smartly and appreciate the role of policymaking and institution-building.

Industry 4.0 shaping the future of work?

In recent years i4.0 has become highly influential in how the FoW is informed at a global scale. By uncovering the ideas driving this digital transformation, we critique what we consider an essentially deterministic view of technology and its implications. In sum, the 'Third Industrial Revolution' appears to have resulted in an intensification of trends already fledgling in the first two: a hollowing-out of employment, a widening distribution of wages and a fall in labour's income share (Andrew Haldane, Chief Economist Bank of England quoted in Dunlop 2016, 111). There is a problem with i4.0 right from the outset: how does the fourth industrial revolution differ from the third?

The debate (limited though it may be) on the Future of Work is dominated by notions of deindustrialisation, dematerialisation and digitalisation. Oxford academics produced the much publicised and widely accepted argument that 47% of jobs in the US were threatened by automation (Frey and Osborne 2017). Tim Dunlop (2016) amongst others has called the approach into question. The Oxford authors fail to make a distinction between jobs and tasks, which clouds their analysis. Thus, where Arntz et al. (2016) took a task-based approach that accounts for the heterogeneity of workers' tasks within occupations, by contrast they estimate that on average across 21 OECD countries just 9% of jobs are automatable. Less commented on is the fact that the Oxford study said jobs *could* be under threat, not that they definitely *would* be.

Industry 4.0 and global value chains

This brings us to a deeper and more fundamental problem with the Oxford debate, whereby it is really only focused on OECD countries, and takes little or no notice of what is happening in the rest of the world (see also Fuchs 2018). There is then a problem of decontextualisation; but more importantly this is fundamentally misleading. For example, the loss of manufacturing jobs in countries such as the US was due as much to their relocation to South East Asia as to technological transformation (see Brenner 2006). And this process was aided by the rise of GVCs with globalisation. Around a third of world trade is now intra-firm, and as much South–South and/or regional as the commonly assumed North–South model.

With i4.0, GVCs also transform. Existing GVCs reshape, taking advantage of the new technologies to re-organise production of existing products or launching new ones (Bianchi and Labory 2018). One tendency that seems to be emerging is reshoring: many companies that have invested in smart manufacturing have reshored production in their home country or in advanced countries (Müller et al. 2017). However, the limitations of reshoring as a driver of a sustainable FoW are identified in specific cases, such as in the UK automotive industry (see Bailey and De Propriis 2014). When factories are organised as networks of robots and cybernetic systems, less labour is needed, and the labour hired in the factory is highly skilled, able to identify and solve problems and able to re-program technology when necessary. Smart manufacturing requires territories where highly skilled labour is available and where infrastructure, particularly connection infrastructure, is sufficiently developed.

New GVCs are also emerging as a result of the development of new products and processes generated from innovation and technological change. Besides reshoring, another trend in production reorganisation that can be outlined is company focus on high phases of the production process, namely pre- and post-manufacturing. Manufacturing is increasingly performed by robots in smart factories that can be located anywhere, provided there is access to energy, high-capacity internet and materials. Key assets of a firm become its knowledge base, its technologies, and its experiences, together with its capacity to identify market trends and consumer tastes, its capacity to innovate and its capacity to renew products and services. As Frederick et al. (2018) conclude in an examination of GVCs in South East Asia:

In the data-driven global economy, companies are expanding their value creation through collaborative ecosystems. Collaboration is a key driver of success in a connected, Industry 4.0

world. It is the convergence of multiple technologies rather than a single technology that, in combination, enables firms to adopt new ways of doing business.

Brun et al. (2019) analyse recent evolutionary trends in GVCs, with a particular focus on the implications of the adoption of the new technologies of the 4IR. They argue that recent dynamics in GVCs include: rationalisation, in that lead firms have tended to reduce the number of suppliers; regionalisation, namely a concentration of production in broad regions, mainly North America, Europe and Asia; and resiliency and digitalisation, which means the use of advanced data analytical tools and physical technologies to improve the digital connectivity and technological capabilities of supply chains.

GVCs and the future of work

A more sanguine approach to the impact of an increasingly GVC/GPN-dominated world economy can be derived from the FoW debate. Quoting the UNCTAD World Investment Report for 2013, Coe and Yeung (2015) argue that:

Some 80 per cent of international trade was now organised through global value chains/production networks coordinated by lead firms investing in cross border production assets and trading inputs and outputs with partners, suppliers and customers worldwide. Empirically, it seems, there remains little doubt that global value chains or global production networks are the most critical organisational platforms through which production in primary, manufacturing and service sectors is coordinated and organised on a global basis.

Furthermore, GPNs reflect relational processes and uneven structures in which and through which corporate power is distributed, exercised and governed (Coe & Yeung, 2015, 65). GPN development has been accelerated and promoted by the move towards global outsourcing (Peck 2017) and the related development of the global logistics industries (Coe 2014). There is now a large and very critical literature on the Apple GVC and in particular Foxconn (see, for example, Clarke and Boersma 2017; Pun et al. 2019). According to the International Labour Organisation's (ILO) *World Employment Social Outlook* (ILO 2015, 136–138), one in five workers globally now works in GVCs, mostly women in emerging economies, driven by the rise of logistics, intermediate goods, wholesale and retail as well as the related fragmentation of the production process. The United Nations report on urbanisation and development (UN-Habitat 2016) argues that growth in informal and vulnerable employment in the rapidly growing cities of the Global South was related to the dynamics of globalisation. The same is true regarding the still enormous numbers of people trapped in various forms of modern slavery. Quoting from the Global Slavery Index, the *Australian Financial Review* Ferguson (2017) points to 45 million slaves globally, with two-thirds in the Asia-Pacific region and 4,300 slaves in Australia through human trafficking, forced labour and servitude. Connecting with GPNs/ GVCs, Ferguson (ibid.) argues that:

Extortion, blackmail, cash back scams and slavery are happening everyday under our noses. It is also happening in the supply chains of businesses either through labour hire companies, or suppliers.

Dyer-Witheyford (2015) argues that the growth of GVCs/GPNs disrupts the notion of a core-periphery. This is reinforced by the ILO's *World Employment Social Outlook* (2015, 39) which suggested that:

In line with increasing FDI coming from developing countries, trade among these countries ('South-South' trade) has been on the rise accounting for almost 5 trillion USD in 2011 and having more than doubled its share in total trade since 1955.

There is one final point to be made about robots, digitalisation, GVCs and the Global South, echoing our argument about the compartmentalisation of the FoW debate, particularly its ethnocentric focus on the countries of the OECD. It has been argued by UNCTAD (Kozul-Wright 2016) and the World Bank Group (2016) that the share of occupations that could experience significant automation is actually higher in developing countries than in more developed areas. Reshoring jobs is one phenomenon that could drive this. For example, Adidas is reshoring production in Germany, developing a so-called 'Speedfactory.' The necessity of adapting to rapidly changing consumer tastes militates against extended supply chains and manual, albeit cheap, labour. Not every job in the Speedfactory will be automated; however, it will create an estimated 160 jobs compared with approximately 1,000 in a typical factory in Asia (The Economist 2017). It is also worth pointing out, given the rise of working- and middle-class demographics in China, that the uptake of robotics in developing economies has been concentrated in China. Furthermore, each year since 2013, China has bought more industrial robots than any other country, and by the end of 2016 was expected to overtake Japan as the world's biggest operator of industrial robots (Kozul-Wright 2016).

Work and productivity: industry 4.0 versus the future of work

One of the differences between FoW and i4.0 is that the latter claims to paint a picture of something new (the 4IR), whilst variations on FoW have reappeared with stunning regularity for decades. The debate about technology and the FoW in successive waves of analysis has witnessed over the last 40 years predictions about a technologically driven meltdown in the world of work that threatens mass poverty and the need for major state intervention in the form of a Universal Basic Income (UBI) (see Rid 2016). Srnicek and Williams (2016: 203, fn 215) point out that fear of automation taking jobs has a long history of which the Luddites were among the earliest examples. Shortly before being assassinated, Martin Luther King, Jr gave a speech about the giant revolutions taking place in the world. There were three elements to the revolution: a human rights revolution; a nuclear weapons revolution; and a technological revolution with the impact of automation and cybernation viewed as a threat to human work creating massive unemployment and soaring inequality.

A long line of works followed Dr King, Jr fore-sighting technology and the FoW. In 1982, Barry Jones published *Sleepers, Wake! Technology & the future of work*. This predicted the potential of new techniques to decimate labour in goods-producing sectors of the economy, an increase in low-skilled service sector work and a computer-driven rise in information-based technologies that further displace human workers (Jones 1982: ix). A short time thereafter, Aronowitz and DiFazio (1994) were pointing to a jobless future and Rifkin (1995) to 'The End of Work.' Brynjolfsson and McAfee (2014) published *The Second Machine Age* which came to three broad conclusions: first, that we are living in a time of astonishing progress with digital technologies at their core; second, that the transformations brought about by digital technology will be profoundly beneficial ones; and third, that digitalisation will bring with it some important challenges which nevertheless can be overcome. But the optimism apparent in this variation to the theme reflects an overall tendency to place excessive stock in the possibilities of technology – particularly when combined with a market-driven allocation of economic resources – to overcome social challenges, despite the short- to medium-term impact of digitalisation and automation that will have very real impacts on levels of unemployment and the sustainability of economic institutions.

The normative theory of a workless future proffered by Brynjolfsson and McAfee and others, in which technology enables abundance for all, has largely avoided criticism to date. An exception is Spencer's (2016, 9) crucial point that overall, the normative view does not factor in that digital technologies reinforce capitalist social relations, with the imperatives of capitalist production to create surplus value placing limits on the development and evolution of digital technologies. It supports the atomistic assumptions of neoclassical economics that technologies are diffused equally across space and time, that all other things remain constant and all information is available to rational agents in the market (Moody 2018). Hence, a second, more pessimistic view is that unless there is significant intervention by governments, such as in the form of a UBI, we can expect a growing threat of revolt from the rapidly expanding ranks of the technologically displaced unemployed. This view is exemplified in Martin Ford's *The Rise of the Robots* (2016), with the author fore-sighting the threat of mass unemployment, induced by AI, as

almost inevitable. Even worse, the concept of Singularity emerges, whereby machine intelligence surpasses human intelligence – leading to a form of cyber serfdom.

Alongside these different views of the FoW, we find Brian Eno (2015), musical innovator *par excellence*, delivering the 2015 John Peel Memorial Lecture for the BBC in the United Kingdom. Eno's approach, which echoes many down the years in the FoW debate, could loosely be described as William-Morris-meets-AI-and-the-digital-revolution, where a distinction is drawn between work and art. Work is what we have to do to survive; all that is left, what we choose to do, is art. Robots will increasingly do the work, so humans can become full-time artists. There are echoes of Marx here – AI and robotics will take over all the work that has to be done, leaving humans to be creative or to focus on work that has intrinsic meaningfulness.

Uniting all these aforementioned approaches to the FOW is a belief in the inevitability that, for large swathes of the population, earning a living from work will no longer be possible. With varying degrees of enthusiasm, and with different models in mind, all (including Eno) come up with some variant of the UBI to provide a safety net through which no-one should fall, and which should provide a standard of living sufficient to provide for (an albeit basic) participation in society. Alternatively, there have been proposals of a 'robot tax', with some modelling suggesting such a tax on accelerated automation can improve wage inequality even as it displaces labour (Zhang 2019). There is then a near-general feeling that mass unemployment (or under-employment) is almost inevitable, and the state will have to intervene extensively to provide the framework that implementation of a UBI would require. So often the overlapping arguments we see here – all converging on the UBI as a panacea to a fully automated future – prescribe a FoW disconnected from material reality and its historical groundings. At a time when the global economy is suffering a long, drawn-out failure to recover in the post-GFC environment, backgrounded by the continued political dominance of the neoliberal paradigm, a UBI might seem as utopian as Brian Eno's version of News from Nowhere. However, these ideas are not new. This is far from being a new proposal, and, as the above account of the history of the FoW debate reveals, is rather one founded in historical precedent.

The hunt for the 'Goldilocks zone' in an industry 4.0-driven future of work

The literature on Industry 4.0 deals with technology, problem-solving of digital transformation issues and embedding firms within production systems. It canvasses issues of standardisation and communication, and making the business and consumer case for the digitalisation of manufacturing production. Attention to the impact of i4.0 on more pressing human dimensions of digitally driven industrial change is significantly lacking. Hence, there is a major concern that the foundation of business model innovation with i4.0 is the mass-scale automation of jobs otherwise performed by humans. Three possible scenarios have been suggested in which i4.0 could create changes to the world of work. These are an automation scenario with digitalised systems directing human actions that largely excludes lower-skilled workers; a hybrid scenario that makes monitoring and control a cooperative process of machine-human interaction but which creates higher demand for human flexibility in response to a range of tasks; and the specialisation scenario in which humans access cyber-physical systems as required, retaining qualified

human work as the central component of a digitally enabled system in which technology becomes a tool to support decision-making (Kotynkova 2017).

Critical perspectives on Industry 4.0 from the social sciences have dissected the technologically deterministic vision that occupies the debate over i4.0 development and its transformative potential. Pfeiffer (2017a) has warned that the technologically deterministic view of i4.0 and its technological marvels threaten a ‘digital despotism’ that will make human workers subservient to integrated digital production systems that monopolise decision-making. It is in this vein that Reischauer (2018) has interpreted Industry 4.0 as a policy-driven discourse intended to embed a digitally driven innovation system in manufacturing production, fulfilling capital’s essential drive to expand in the form of a digitally driven regime of accumulation. Yet these critical inroads on the human organisational dimensions of i4.0 remain on the fringe of i4.0 research and policymaking. The focus of most research is on macro-level changes to manufacturing production and a technological determination of outcomes is being produced in what Morgan (2019, 14) defines as a ‘quasi-determinism’ where ‘anxiety combined with passivity and complacency are being produced.’ The foundation provided by technical and business approaches to interpreting i4.0 is easily transferrable from the template that global management consulting firms mobilise to provide techno-centric solutions to manufacturing firms and corporations for implementing automation strategies and raising employee productivity (see BCG Perspectives 2017; Bechtold et al. 2014; McKinsey 2015). The innovations proposed by Industry 4.0 that they commonly cite may imply significant opportunities for improving manufacturing output, market share and competitiveness, and for driving business model innovation amongst manufacturing firms. Ultimately, these drives the transfer of recommendations to the policy responses of governments throughout the world. But according to Kotynkova (2017), ultimately income and social inequality will loom large as those with the capital benefit most from automation, whilst the millions of people dependent on manufacturing work are dispossessed of their livelihoods by smart machines.

To show how little such considerations have been factored into the perspective of i4.0’s most vocal proponents, Pfeiffer (2017b) has also interpreted it as a marketing exercise that provides an imaginary vision of a future in which none of the sociological implications have been given much detail. Fuchs (2018) has echoed this sentiment, arguing that Industry 4.0 acts as a ‘capitalist panacea’ to economic and social problems that ignore the existing social and economic inequalities being created by the ‘disruptive’ nature of digital capitalism. As a result, the abovementioned scenario of specialisation represents a potential ‘Goldilocks zone’ fix for a FoW in which i4.0 is positively transformative (Seet et al. 2018). However, at present, such an outcome of i4.0 appears far beyond the scope of its development given dominating neoliberal political-economic conditions. In the following section, we outline the expression of i4.0 that has most visibly, empirically impacted labour markets, production and possible trajectories for the FoW.

Industry 4.0 and the future of work: digital platform capitalism

The search for the Goldilocks solution in terms of work and employment for both i4.0 and the mainstream FoW approaches seems to be somewhat utopian. Instead, we now turn to the character, form and function of what has come to be called ‘Platform Capitalism,’

a more critical approach to the emerging world supposedly driven by i4.0 or varieties of FoW, and examine the implications of the emerging digital economy for organisational structures, work and employment. Wu and Gereffi (2018, 351) argue that 'in terms of whether the Internet era has transformed the basic governance structure of global value chains, the evidence is conclusive that a platform economy now exists'. The mainstream debate is relatively silent on the political-economic dimensions of the FoW so fails to address technological change in its human dimensions and respond to its negative effects. In relative terms, the consequences of a pervasive techno-centric view of change is most evident where capitalism's instability is a burden largely shouldered by workers in an expanding 'gig economy.'

Digital platforms are causing significant levels of disruption to the world of work and challenging traditional business models, driven in large part by a tendency towards monopoly. Of course, centralisation and concentration are essential features of capitalism. What is different this time is the speed with which new winners achieve enormous scale and with relatively small direct employment. A key reason for the speed of digital capitalism's colonisation of the FoW is that platforms represent a business model based on the control of data. The monopoly characteristics of platform capitalism express the most widespread application of digital technologies. The rhetoric around the sharing economy is less a movement for more openness than it is for deregulation. Slee (2017, 19) reminds us that '[i]t's not about building an alternative to a corporate-driven market economy, it's about extending the deregulated free market into new areas of our lives.' As a response to the new monopolists – 'smiley faced' monopolists – as the *New Internationalist* has compared them with the robber barons of the interwar period (Baird 2016), the *Financial Review* has suggested that there may be a need for rethinking monopoly and anti-trust laws (Foroohar 2017). This is certainly essential to a sustainable FoW given how technology facilitates the expansion of globally financialised monopoly capital.

With platform capitalism, the tendency towards monopoly is pronounced, but that does not mean that competition disappears, simply that it changes form. The new giants are faced with an increasingly competitive environment comprised of other platforms (ibid., 97). Thus, for capitalism to function on digital platforms, the vast amounts of data that humans create become resources. Srnicek and Williams (2016, 48) contend that in acting as a new type of firm, platforms are able to extract and control data by providing the software and hardware essential to mediating between users, the increased use of these platforms becoming a means to create network effects that generate open-source material and increase the platform's value, hence its monopoly power. Although Srnicek (ibid.) identifies other types of digital platform (advertising, cloud, industrial, product and lean), they are unified in twenty-first century capital's use of data as a new raw material. And each type of platform holds within it the distinct ability to discipline workers in new and potentially more invasive ways. The basis of new employment on digital platforms is growing at such a rapid pace that the industrial relations laws and regulations of many countries are lagging, spelling negative consequences for vulnerable workers. With the rapid pace of technological change, it is not beyond reason to imagine how the application of digital platform business models in manufacturing plants in the developed world could find high-skilled employees joining a growing pool of workers exposed to the gig economy and the precarious conditions, insecurity and low pay that accompanies it (see

Dean and Spoehr 2018). This is a predicament facing the manufacturing sector that very little academic work is yet to address beyond the predicted regulatory challenges produced by phenomenal digitally driven debasement of high-skilled, high-waged work (see De Ruyter et al. 2019).

Platform capitalism and GVCs: implications of industry 4.0 in the future of work

In an extensive report prepared for the ILO, De Stefano (2016) looked at the rise of a 'just-in-time' workforce made possible with digitalisation's creation of a gig economy characterised by on-demand work, crowd work and weakened labour protections, drawing a distinction between crowd work and work-on-demand via apps. However, both allow for far-reaching personal outsourcing of activities to individuals rather than complex businesses. Both forms rely on and reinforce the fragmentation, commodification and isolation of labour that has been a characteristic of the neoliberal era. This allows for new forms of flexibility through humans as-service (Ibid., 4). The 'work' nature of this labour is concealed and workers themselves are invisible to all intents and purposes. Risk and responsibility are handed down to individual workers. In this sense, the 'gig economy' reflects the worst aspects of the jobbing musician's life (De Ruyter and Brown 2019), and is simply another facet of a much vaster trend towards casualisation of labour, as we have seen (De Stefano 2016, 6). De Stefano (Ibid.) concludes that:

The gig economy thus is not only an extreme form of the fissuration of business organisation, but is in turn affected by the same fragmentation of workplaces, and the related multiplication of centres, as other sectors of the economy. Arguably, indeed, platforms and apps may carry out, in some cases, the activities that private employment agencies execute in other sectors, normally without being subject to the systems of licensing and regulation of these agencies.

Some people may benefit from the 'flexibility' of platform capitalism, but the reality of working life in the 'gig economy' of the creative industries so beloved by proponents of platform capitalism is simply tough. Indeed, platform capitalism is predicated on, and accelerates tendencies towards, the individualised, atomised, unprotected, risky and uncertain labour market which the neoliberal ascendancy has sought so assiduously to promote. Montalban et al. (2019) conclude that the 'new' wage-labour nexus institutionalised by platform capitalism, particularly the 'Uberisation of labour,' is not as recent as it would appear. It is a more radical version of certain existing processes. There is also a self-reinforcing relationship between inequalities and the development of the platform economy. Poverty and inequalities encourage the development of platforms and platforms, in turn, may increase poverty by the dis-empowerment of low-skilled labour and the rise of very high-paid 'stars.' Srnicek and Williams (2016, 104) argue that the extensive automation of work would produce seven distinct trends in the years to come. They suggest that globalisation and automation will lead to intensified working-class precarity, and urban marginality in developed countries; jobless economic recoveries; growth in slum populations; transformation of higher education into job training to match demand for high-skill workers; slow economic growth; and changes to immigration, incarceration and benefits schemes within territories as those made technologically unemployed are made subject to coercive controls and survival economies.

Zuboff (2019) takes the concept of 'platforms' to an even higher stage, which she calls 'surveillance capitalism'. This is based on the argument that companies, 'the platforms' such as Google, have discovered a way to translate non-market interactions with users into surplus raw materials for the fabrication of products aimed at their real customers – advertisers. This is the behavioural surplus. These surveillance assets become surveillance revenues which are then translated into surveillance capitalism:

Industrial capitalism transformed nature's raw materials into commodities, and surveillance capitalism lays its claim to the stuff of human nature for a new commodity invention ... the essence of exploitation here is the rendering of our lives as behavioural data for the sake of others' improved control of us (Zuboff, *ibid.*, 94).

Conclusion

We have come a long way from Industry 4.0 to move towards the beginning of a theoretical explanation for why it is becoming a buzzword in the academic literature, having already embedded itself in the lexicon of global consulting business and the policies of numerous governments. In this paper, we have sought to theorise the impact of a so-called 4IR in spatial and material dimensions. Specifically, it is arguable that i4.0 differs only incrementally from the technologies developed during the third industrial revolution, with the emergence of the internet – and its proliferation in an Internet of Things – aiding in the contemporary era an industrial IoT that facilitates the integration of digital technologies with implications for manufacturing production. Industry 4.0 appears to have the greatest implications for a much greater integration and coordination of Global Value Chains and their role in the reorganisation of capitalist production, already spread widely across the global economy, into the newly prepared terrain of cyberspace. Herein there is taking place a veritable gold rush to commodify data via the opening of another front in the spatio-temporal expansion of the global capitalist regime of accumulation.

The combination of i4.0 with GVCs and a global organisation of production has significant implications for the future of quality work at a global scale. Our theoretical development has concluded with the notion that the first wave of i4.0's colonisation of cyberspace can be seen in 'platform capitalism.' Where digitalisation both disrupts and obscures digital work, the use of digital technologies to do this implies the ability of i4.0's digitally integrated technologies to 'piggyback' on the connectivity of manufacturing production through GVCs, and potentially produce for manufacturing workers in the Global North disruptions similar to those already experienced by workers in the Global South impacted most of all by the effects of GVCs.

Within a neoliberal political economy, critical investigations point towards a technologically neutral embrace of digital technologies and the 4IR. Beyond this, however, is the potential for i4.0 to underscore a digitally driven transformation of our social institutions. At present, digital platforms ranging from Amazon to Uber to Deliveroo immiserate millions across the world thrust into such insecure, precarious forms of work now rendered digital, on-demand and destructive to established patterns of quality, human work. Contemporary global political-economic conditions could exacerbate this. Only through the integration of digital technologies with human work to augment our labour and where possible, help drive societies and economies away from the patterns of

commodification already underpinning digital platforms, can digitalisation produce a high-quality globally inclusive Future of Work.

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